Lecture 3

Consumption, Saving, Interest Rates

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Today: your first dynamic modern macro model

- So far: static models = models without time
- But lots of questions in macro are inherently dynamic = they involve time
  - Why do economies grow? Why $\approx$ constant long-run growth rates?
  - Why booms, busts? Should policy stabilize business cycles? How?
  - In typical recession, why does I decline more than C?
  - even true for questions we've already covered, e.g. productivity increased over time $\implies$ reduction in average hours worked
  - ... but so far we discussed such issues "outside of the model"

Today: your first dynamic modern macro model

- **Today and rest of course:** there is time $t = 0, 1, 2, ...$ (typically discrete)
- May have already seen a dynamic model = Solow model, but no microfoundations (so $\neq$ modern macro)
- Also recall Keynesian IS-LM model from Econ 101: t-subscripts but not really a proper dynamic model. And definitely not microfounded.

Plan

1. The Keynesian view of consumption
2. A two-period model of consumption
3. The permanent income hypothesis
4. Consumption, saving and interest rates in general equilibrium
5. Consumption, asset prices and firm value

Background Readings

- Follow closely Kurlat Chapter 6
- Further reading for the asset-pricing extension: Altug and Labadie (2008), Chapter 8, pp. 162–173.

The Keynesian view of consumption

Keynes on consumption

- Keynes in his 1936 "General Theory of Interest, Employment & Money": "The fundamental psychological law, upon which we are entitled to depend with great confidence both a priori from our knowledge of human nature and from the detailed facts of experience, is that men are disposed, as a rule and on the average, to increase their consumption as their income increases but not by as much as the increase in the income."

- In equations, consumption C depends on income Y as

$$C = c(Y) \quad \text{with} \quad c'(Y) > 0 \quad \text{and} \quad c'(Y) < 1$$

- $c(Y)$ = consumption function. Question for you: name for $c'(Y)$?

- Depending on how interpret language, he may also mean

$$\frac{c'(Y)Y}{c(Y)} = \frac{\partial \log c(Y)}{\partial \log Y} < 1$$

  i.e. elasticity of C with respect to $Y < 1$

Importance of MPC for multipliers (Keynesian cross)

Note: MPC $= \gamma$, consumption function $C = \alpha + \gamma(Y - T)$

Keynesian Cross

- Equilibrium condition in simple Keynesian Cross model
  - Output must equal planned expenditures:

$$Y = \alpha + \gamma(Y - T) + I + G + NX$$

- A little bit of algebra then gives:

$$Y = \frac{1}{1-\gamma}[\alpha - \gamma T + I + G + NX]$$

- Government purchases multiplier $= \frac{1}{1-\gamma}$
- Tax cut muliplier $= \frac{\gamma}{1-\gamma}$

In micro data both $c'(Y)$ and $\frac{\partial \log c(Y)}{\partial \log Y}$ are considerably < 1

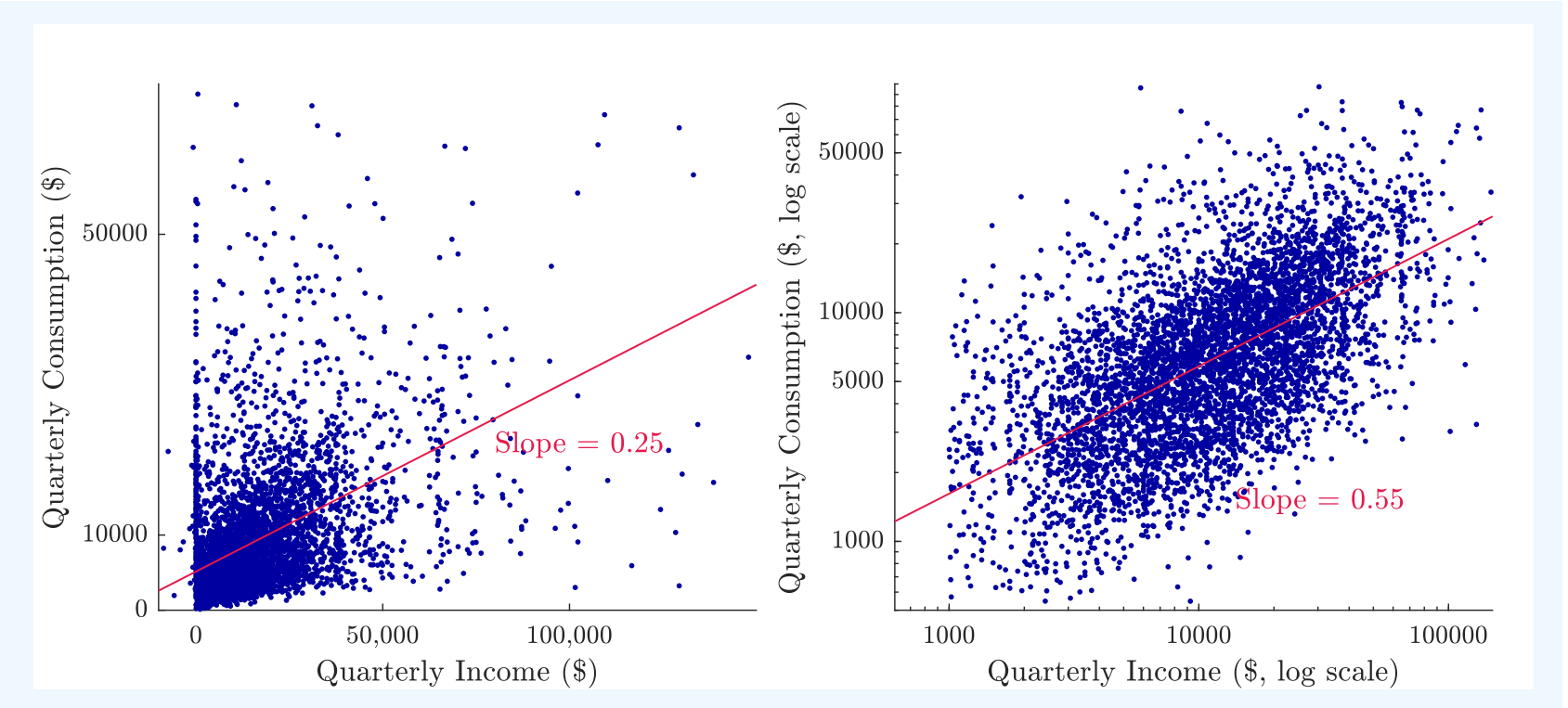


Fig. 6.1.1: *Evidence on the Keynesian consumption function. Each dot represents a household. Source: Consumer Expenditure Survey, 2014.*

Implications for aggregate consumption and income

- Recall: in micro data both $c'(Y)$ and $\frac{\partial \log c(Y)}{\partial \log Y}$ are considerably < 1
- Around 1950s: seen as conclusive evidence in favor of Keynesian consumption function
- But then what happens as countries grow richer over time and Y grows?
- If $\frac{\partial \log c(Y)}{\partial \log Y} < 1$ then we should have

$$\frac{c(Y)}{Y} \quad \text{declining with } Y$$

  so people consume proportionately less and less as country grows richer

- Do we see this? No.

In macro data $\frac{\partial \log c(Y)}{\partial \log Y}$ is much larger, $\frac{C}{Y}$ stable over time

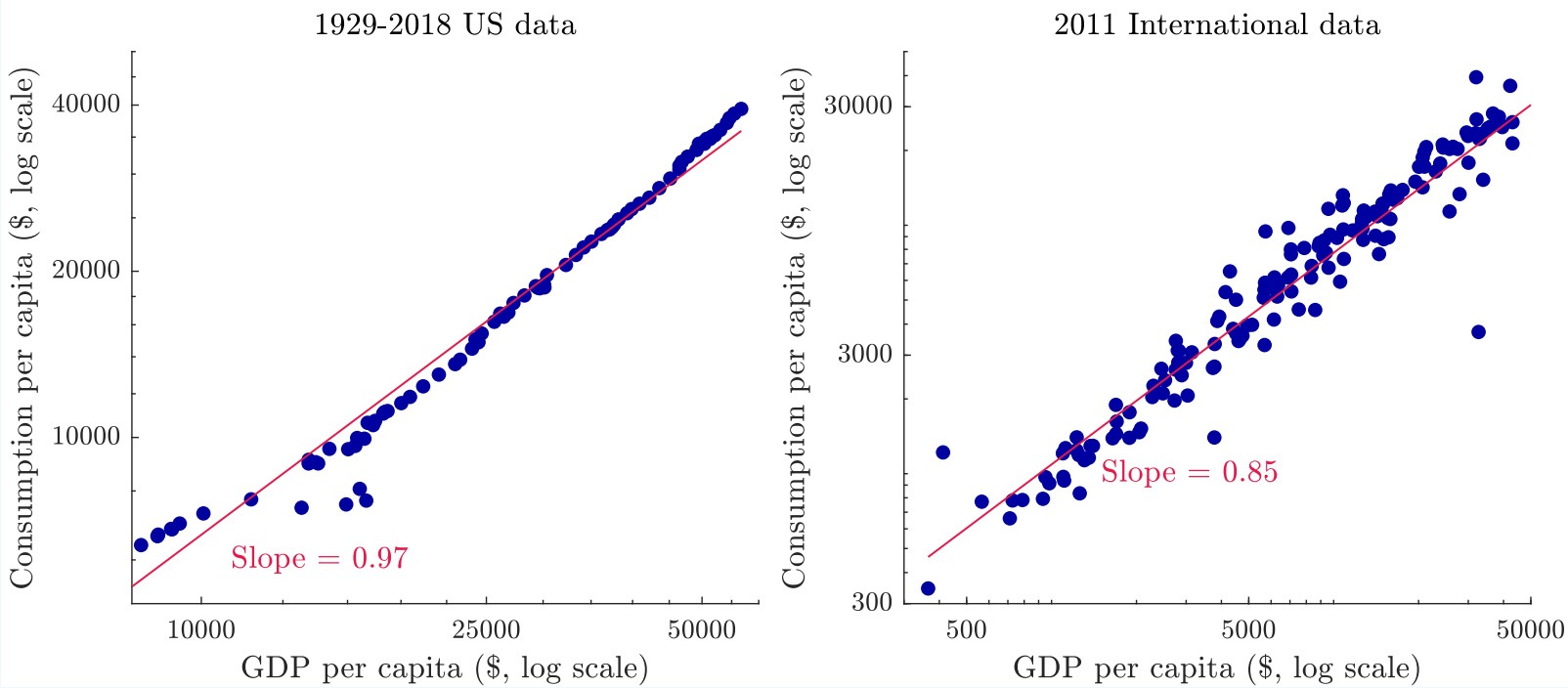


Fig. 6.1.2: *Evidence on the Keynesian consumption function from aggregate data. The left panel is US time-series evidence; the right panel is cross-country evidence. Sources: NIPA and Feenstra et al. (2015)*

Summary

- In micro data both $c'(Y)$ and $\frac{\partial \log c(Y)}{\partial \log Y}$ are considerably < 1
- but in macro data, $\frac{\partial \log c(Y)}{\partial \log Y}$ is much larger (perhaps even ≈ 1), $\frac{C}{Y}$ stable over time
- Seems puzzling, what's going on?
- Next: use economic theory to resolve puzzle

A two-period model of consumption

A two-period model of consumption

- For now: study problem of individual in isolation taking as given prices (partial equilibrium)
- In a bit: general equilibrium
- Two time periods $t = 1$ and $t = 2$
- Consumption c_1 and c_2, income y_1 and y_2
- Utility function

$$u(c_1) + \beta u(c_2)$$

with u strictly increasing, concave, discount factor $0 < \beta < 1$

- Vocabulary that sometimes comes up: discount factor vs discount rate
  - discount factor: β, typically a bit below one
  - discount rate: ρ, typically a bit above zero
  - link: $\rho = 1/\beta - 1 \Leftrightarrow \beta = \frac{1}{1+\rho}$, e.g. $\beta = 0.95, \rho = 1/0.95 - 1 = 5.26\%$

A two-period model of consumption

Household solves

$$\max_{c_1, c_2, a} \; u(c_1) + \beta u(c_2) \quad \text{s.t.}$$

$$c_1 + a = y_1 \tag{1}$$

$$c_2 = y_2 + (1 + r)a \tag{2}$$

Notation:

- c_1, c_2: consumption at $t = 1$ and $t = 2$
- y_1, y_2: income at $t = 1$ and $t = 2$
- r: interest rate (for now exogenously given)
- a: saving
- Note: a can be negative, $a < 0$ means household is borrowing
- From (1) $a < 0 \Rightarrow c_1 > y_1$, i.e. consume more than income by borrowing

Formulation in terms of present-value budget constraint

- Implicit assumption: can borrow and save as much you want at rate r
- then can combine (1) and (2) into present-value budget constraint

$$c_1 + a = y_1 \quad \text{and} \quad a = \frac{c_2}{1+r} - \frac{y_2}{1+r} \quad \Rightarrow \quad \underbrace{c_1 + \frac{c_2}{1+r}}_{\text{PV of consumption}} = \underbrace{y_1 + \frac{y_2}{1+r}}_{\text{PV of income}}$$

  where PV stands for present value.

- Hence households maximize utility s.t. present-value budget constraint

$$\max_{c_1, c_2} \; u(c_1) + \beta u(c_2) \quad \text{s.t.}$$
$$c_1 + \frac{c_2}{1+r} = y_1 + \frac{y_2}{1+r}$$

Euler equation

- Optimality condition

$$u'(c_1) = \beta(1+r)u'(c_2) \qquad (*)$$

- Intuition: consume \$1 now vs save and consume later
  - consume \$1 now: utility increases by $u'(c_1)$
  - save \$1: have $\$(1+r)$ at $t=2$, utility by $\beta u'(c_2)$ for each \$

- $(*)$ is called "Euler equation" after Leonard Euler, plays an important role in macroeconomics

- Note: "Euler equation" sounds fancy but simply means "intertemporal optimality condition"

Graphical representation

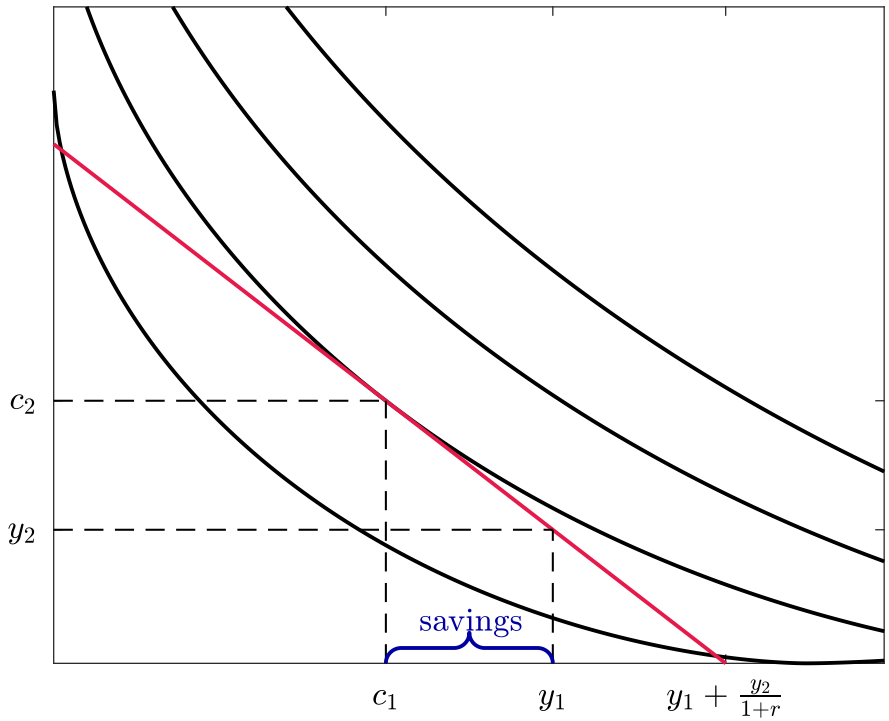
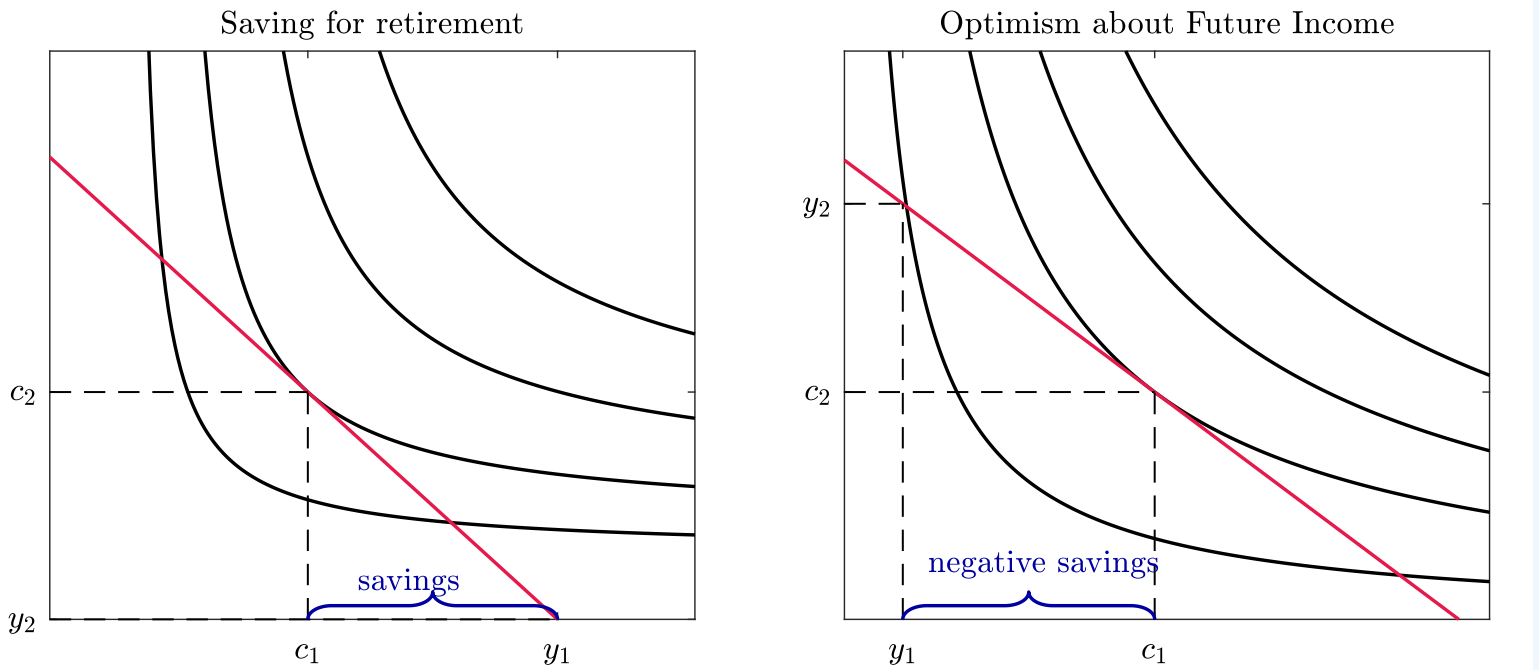


Fig. 6.2.1: *The consumption-savings decision as a two-good consumption problem.*

A saver (left panel) and a borrower (right panel)



Left panel: $y_2 = 0$ and $y_1 > 0 \Rightarrow$ save. Right panel: $y_1 << y_2 \Rightarrow$ borrow.

What happens when r changes?

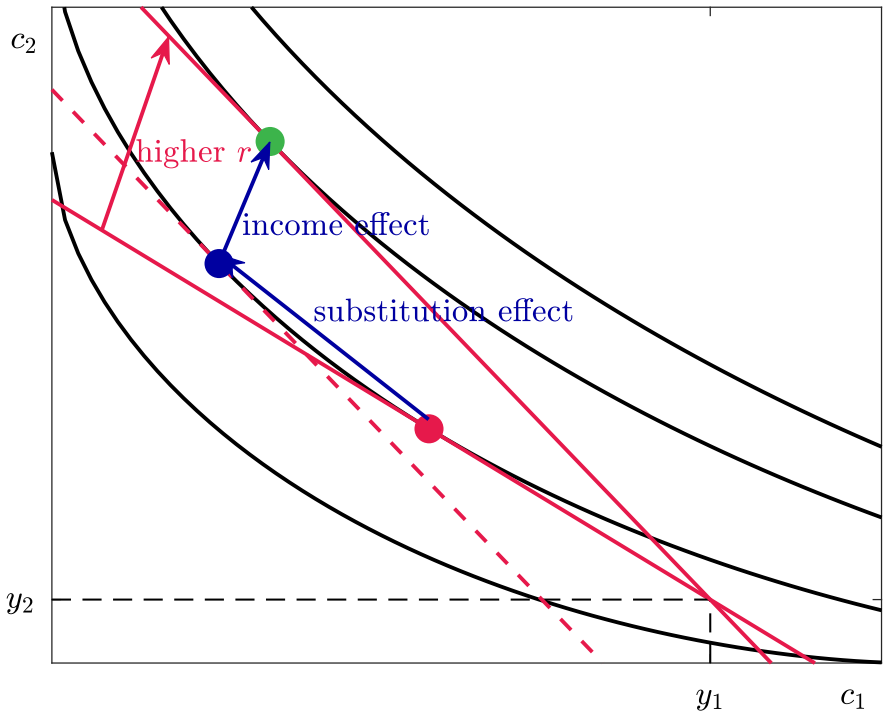
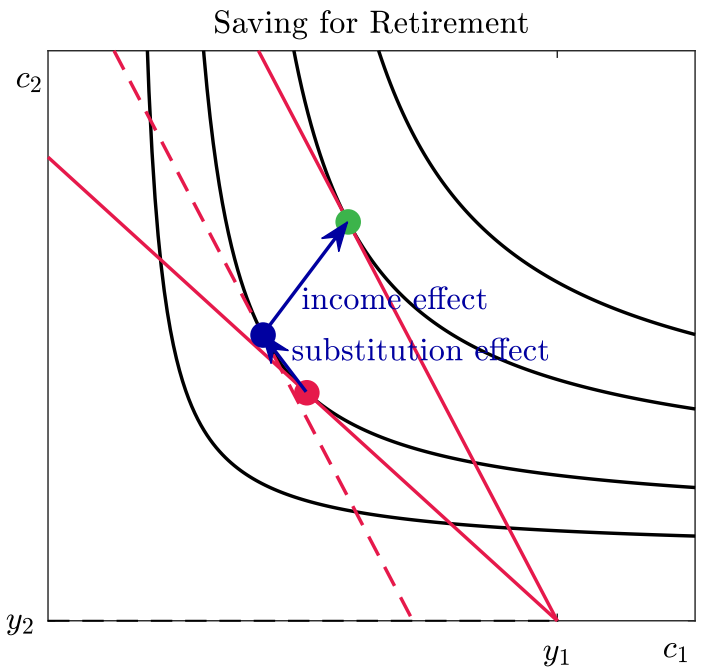
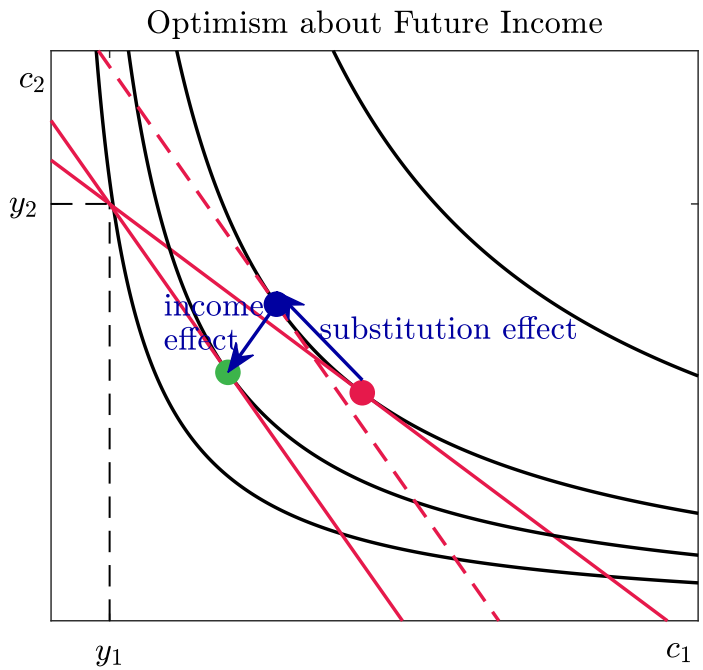


Fig. 6.2.3: *Consumption response to higher interest rates.*

Effect of change in r for saver (left) and borrower (right)





The permanent income hypothesis

An important observation about our model

- Can write two-period model as

$$\max_{c_1,c_2} \; u(c_1) + \beta u(c_2) \quad \text{s.t.} \quad c_1 + \frac{c_2}{1+r} = W$$

$$\text{where} \quad W = y_1 + \frac{y_2}{1+r} = \text{PV of income}$$

  sometimes also called "lifetime income" or "permanent income"

- Can immediately see: c_1 will depend on y_1, y_2 only through W, i.e.

$$c_1 = c(W), \quad W = y_1 + \frac{y_2}{1+r}$$

- It's not current income that matters, but PV of current + future income
- This way of thinking: "permanent income hypothesis" (Friedman, 1957)
- Contrast with Keynes quote, captured w consumption function $c_1 = c(y_1)$

- Can already see: our model is really very different from Keynes view
- Next: flesh this out with parametric example

A parametric example

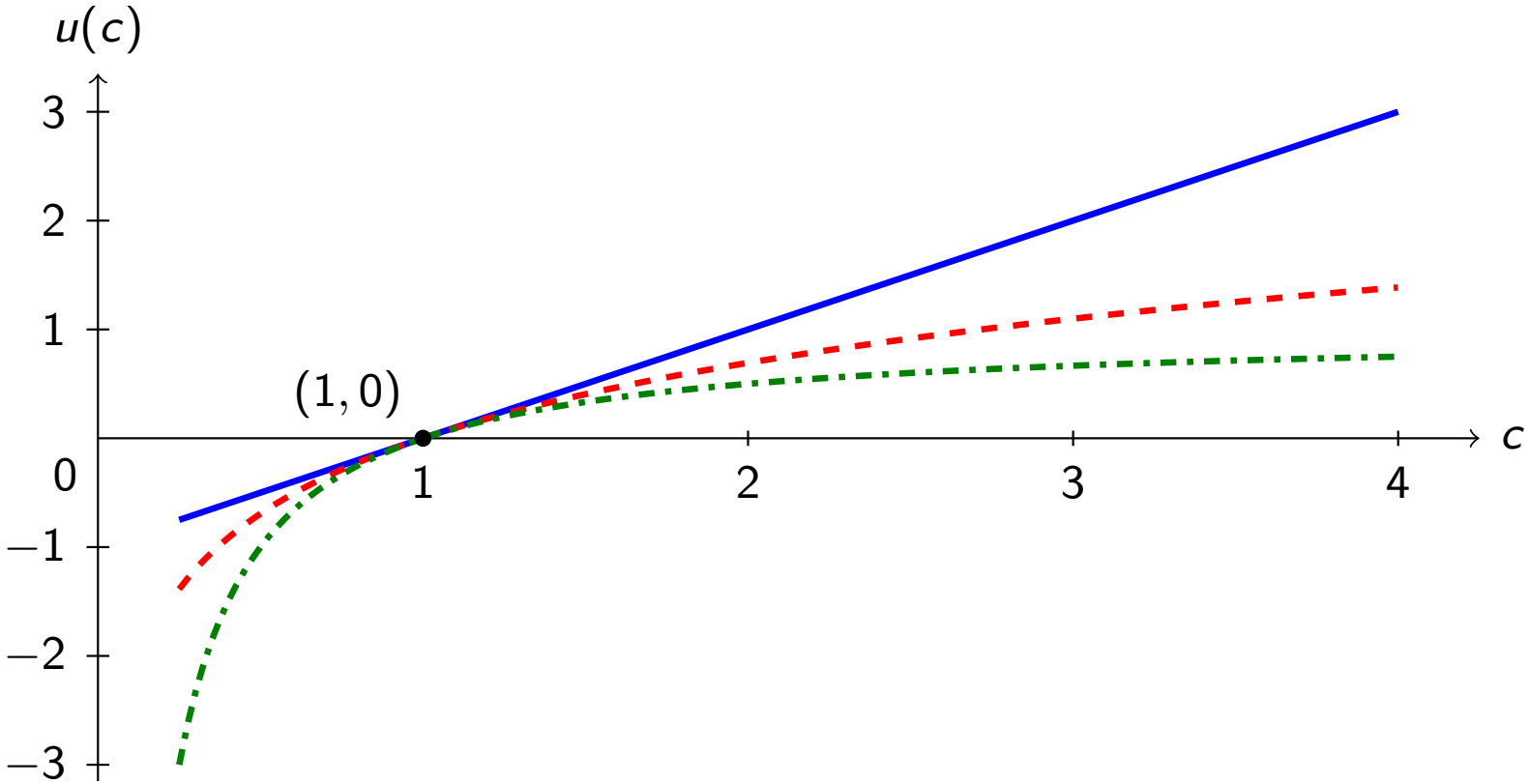
$$u(c) = \frac{c^{1-\sigma} - 1}{1 - \sigma} \qquad (**)$$

- $1/\sigma$ is called "intertemporal elasticity of substitution (IES)"
- Sometimes σ is called "coefficient of relative risk aversion (RRA)" and $(**)$ is called "CRRA utility"
- Can show: log utility = special case with $\sigma = 1$ (use l'Hopital's rule)

$$u(c) = \lim_{\sigma \to 1} \frac{c^{1-\sigma} - 1}{1 - \sigma} = \log c$$

- Note: weird -1 in numerator only there because we want to take this limit

CRRA utility: three examples



———	$\sigma = 0:$	$u(c) = c - 1$	Linear utility
- - -	$\sigma = 1:$	$u(c) = \log c$	Log utility
-·-·-	$\sigma = 2:$	$u(c) = 1 - 1/c$	Higher relative risk aversion

Euler equation in parametric example

- With functional form $(**)$ have

$$u'(c) = c^{-\sigma}$$

- Euler equation

$$u'(c_1) = \beta(1+r)u'(c_2) \quad \Rightarrow \quad c_1^{-\sigma} = \beta(1+r)c_2^{-\sigma}$$

or

$$\frac{c_2}{c_1} = [\beta(1+r)]^{1/\sigma}$$

- IES $1/\sigma$ governs responsiveness of growth rate of consumption c_2/c_1 to changes in r and β, e.g.

$$\frac{\partial \log(c_2/c_1)}{\partial \log(1+r)} = 1/\sigma \quad \Rightarrow \text{hence the name IES}$$

- A low IES $1/\sigma$ means households dislike intertemporal substitution (want to smooth c) $\Rightarrow$ low responsiveness of c_2/c_1 to changes in r and β

Analytic solution in parametric example

- Not hard to show (you should be able to!): solution to problem is

$$c_1 = \frac{\left(\frac{1}{\beta(1+r)}\right)^{1/\sigma}(1+r)}{1+\left(\frac{1}{\beta(1+r)}\right)^{1/\sigma}(1+r)}W, \quad c_2 = \frac{1+r}{1+\left(\frac{1}{\beta(1+r)}\right)^{1/\sigma}(1+r)}W$$

  where

$$W = y_1 + \frac{y_2}{1+r}$$

- Useful special case (see next slide why): $\beta(1+r) = 1$

$$c_1 = c_2 = \frac{1+r}{2+r}W$$

- If further $r = 0$, then $c_1 = c_2 = \frac{W}{2}$

Reasonable parameterizations

Reasonable parameterizations have two features

1. $\beta(1+r)$ not too far from 1, equivalently discount rate $\rho = 1/\beta - 1 \approx r$
   - e.g. $\rho = 0.05, r = 0.01$ but not $\rho = 0.5, r = 0.01$ or $\rho = 0.05, r = 0.5$
   - in general equilibrium model of part 4, $1 + r^* \approx 1/\beta$
   - market discounts future at roughly the same rate as individuals
2. σ not too small
   - e.g. $\sigma = 2$, probably ≥ 1, definitely not $\sigma = 1/5$ or $1/10$
   - empirical evidence: people do not massively substitute intertemporally, i.e. they do not massively increase spending when $r \downarrow$

MPC out of transitory income shock

- Question: suppose y_1 increases by \$1 but y_2 is unchanged. By how much does c_1 increase? What is MPC out of transitory income shock?

- Answer: the marginal propensity to consume (MPC) out of a transitory income shock is

$$\frac{\partial c_1}{\partial y_1} = \frac{\left(\frac{1}{\beta(1+r)}\right)^{1/\sigma}(1+r)}{1+\left(\frac{1}{\beta(1+r)}\right)^{1/\sigma}(1+r)}$$

- Contrast with Keynesian cross: $C = \alpha + \gamma(Y - T)$, $\gamma =$ exogenously given

- In current model, MPC is instead endogenous and depends on preferences (β, σ) and prices (r)!

- This is precisely what we mean when we say "Keynesian cross is not microfounded but modern macro models are"

MPC out of transitory income shock

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- Answer: the marginal propensity to consume (MPC) out of a transitory income shock is

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- Also note: this MPC is a number < 1 and $<< 1$ for reasonable parameters

- In fact, in this 2-period model, MPC $\approx 1/2$, e.g. with $\beta(1+r) = 1$

$$\frac{\partial c_1}{\partial y_1} = \frac{1+r}{2+r}$$

and if $r = 0$ then $c_1 = W/2$ and

$$\frac{\partial c_1}{\partial y_1} = \frac{1}{2}$$

MPC out of permanent income shock

- Question: suppose both y_1 and y_2 increase by \$1. By how much does c_1 increase? What is MPC out of permanent income shock?

- Assumption: $y_1 = \bar{y}_1 + \Delta, y_2 = \bar{y}_2 + \Delta$ and Δ increases

- From $W = y_1 + \frac{y_2}{1+r}$: when both $y_1, y_2 \uparrow$ by \$1, $W \uparrow$ by $\approx$ \$2

$$\frac{\partial W}{\partial \Delta} = 1 + \frac{1}{1+r} = \frac{2+r}{1+r} \approx 2$$

- Hence

$$\frac{\partial c_1}{\partial \Delta} = \frac{\partial c_1}{\partial W}\frac{\partial W}{\partial \Delta} = \frac{\left(\frac{1}{\beta(1+r)}\right)^{1/\sigma}(2+r)}{1+\left(\frac{1}{\beta(1+r)}\right)^{1/\sigma}(1+r)}$$

- For reasonable parameters this is a number ≈ 1. Exact when $\beta(1+r) = 1$

$$\frac{\partial c_1}{\partial \Delta} = 1$$

Summary: main prediction of permanent income hypothesis

- Main prediction of permanent income hypothesis (PIH):

  MPC out of transitory income $<<$ MPC out of permanent income

- Intuition: save large fraction of transitory income change so as to smooth consumption over time

Extension to many time periods

- Macro models used for research, policy etc: not just two time periods $t = 1, 2$. Instead: many time periods $t = 1, 2, 3, ..., T$
  - consumption $c_1, c_2, c_3, ..., c_T$, income $y_1, y_2, y_3, ..., y_T$
  - relatively straightforward extension – see Kurlat chapter 6.3
- Takeaway: more time periods make PIH prediction even starker
- Key: PV budget constraint

$$c_1 + \frac{c_2}{1+r} + \frac{c_3}{(1+r)^2} + ... + \frac{c_T}{(1+r)^{T-1}} = W$$

$$\text{where} \quad W = y_1 + \frac{y_2}{1+r} + \frac{y_3}{(1+r)^2} + ... + \frac{y_T}{(1+r)^{T-1}}$$

- Makes PIH prediction even starker: effect of transitory change in y_1 (say) has negligible effect on permanent income W, hence $\partial c_1 / \partial y_1$ very small
- Example: if $\beta(1+r) = 1$ and $r = 0$, then $c_1 = c_2 = ... = W/T$ and

$$\frac{\partial c_1}{\partial y_1} = \frac{1}{T} = \text{small number}$$

Friedman's example: paydays staggered throughout week

CHAPTER IX

Summary and Conclusion

THE central theme of this monograph can be illustrated by a simple hypothetical example. Consider a large number of men all earning $100 a week and spending $100 a week on current consumption. Let them receive their pay once a week, the pay days being staggered, so that one-seventh are paid on Sunday, one-seventh on Monday, and so on. Suppose we collected budget data for a sample of these men for one day chosen at random, defined income as cash receipts on that day, and defined consumption as cash expenditures. One-seventh of the men would be recorded as having an income of $100, six-sevenths as having an income of zero. It may well be that the men would spend more on pay day than on other days but they would also make expenditures on other days, so we would record the one-seventh with an income of $100 as having positive savings, the other six-sevenths as having negative savings. Consumption might appear to rise with income, but, if so, not as much as income, so that the fraction of income saved would rise with income. These results tell us nothing meaningful about consumption behavior; they simply reflect the use of inappropriate concepts of income and consumption.

Source: Friedman (1957) "A Theory of the Consumption Function"

`https://www.nber.org/books-and-chapters/theory-consumption-function`

This excerpt from conclusion

`https://www.nber.org/system/files/chapters/c4411/c4411.pdf`

The PIH predicts that multipliers are small

Keynesian Cross

- Equilibrium condition in simple Keynesian Cross model
  - Output must equal planned expenditures:

$$Y = \alpha + \gamma(Y - T) + I + G + NX$$

- A little bit of algebra then gives:

$$Y = \frac{1}{1-\gamma}[\alpha - \gamma T + I + G + NX]$$

- Government purchases multiplier $= \frac{1}{1-\gamma}$
- Tax cut muliplier $= \frac{\gamma}{1-\gamma}$

PIH predicts that aggregate MPC out of current income is small, say $\gamma = 0.05$

$\Rightarrow$ Multipliers are small

- government purchases multiplier ≈ 1, say $= \frac{1}{1-0.05} = 1.053$
- tax cut multiplier close ≈ 0, say $= \frac{0.05}{1-0.05} = 0.053$

Resolving our puzzle with the permanent income hypothesis

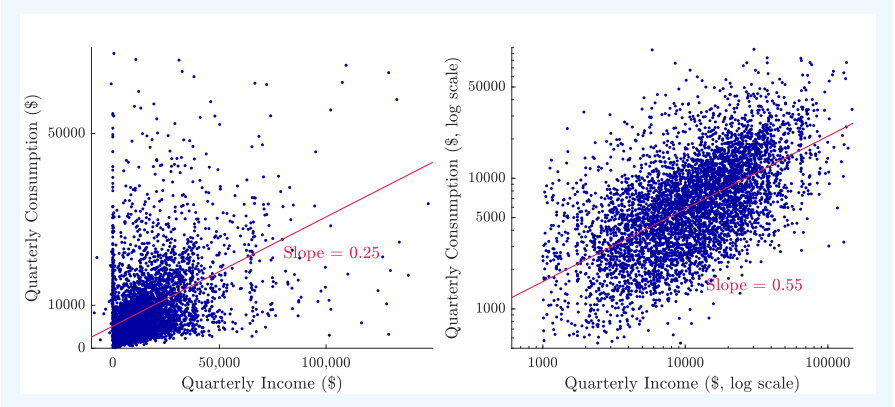


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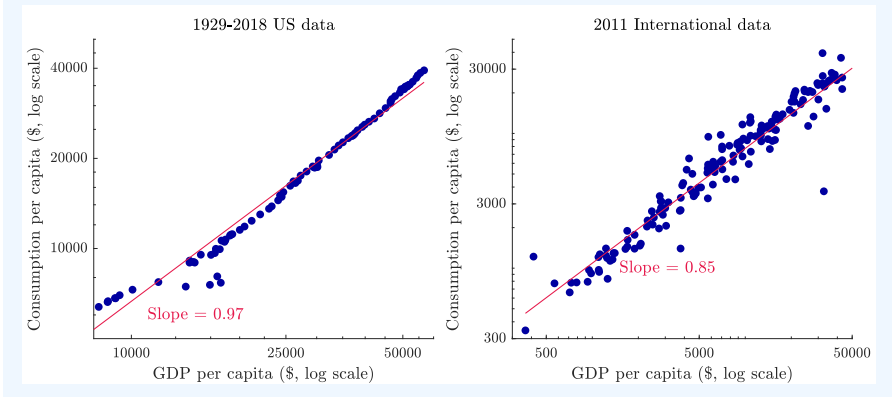


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Resolving our puzzle with the permanent income hypothesis

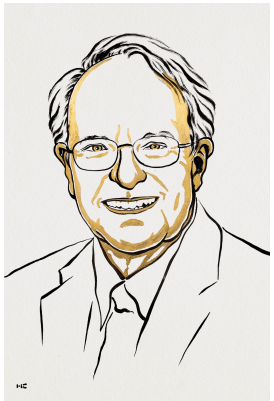
Micro data:

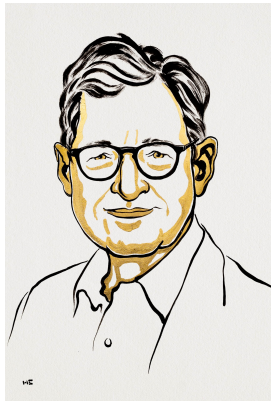
- Transitory income shocks are important, some people have good year, others have bad year. Get mixed together with permanent income shocks.
- Coefficient in regression of C on Y reflects MPCs out of both permanent and transitory income shocks
- Estimate $c'(Y)$ and $\frac{\partial \log c(Y)}{\partial \log Y}$ considerably < 1

Macro data:

- Transitory income shocks mostly wash out in the aggregate
- Coefficient in regression of C on Y mostly reflects MPCs out of permanent income changes (e.g. country as a whole is richer on average)
- Estimate $c'(Y)$ and $\frac{\partial \log c(Y)}{\partial \log Y}$ much closer to 1

The 2026 Nobel Prize: Hall and Woodford





Robert E. Hall **Michael Woodford**

Illustrations: Niklas Elmehed © Nobel Prize Outreach

Announced 12 October 2026

"for microeconomic foundations of macroeconomic stabilisation"

- Modern macroeconomics: decisions today depend on the future.
- Woodford: monetary policy through interest rates and expectations.

- Our focus: Hall, consumption smoothing and the Euler equation.

Sources: NobelPrize.org; Cochrane (2026), *A Great Nobel*.

Hall (1978): turning consumption smoothing into a test

Stochastic Implications of the Life Cycle–Permanent Income Hypothesis: Theory and Evidence

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Robert E. Hall

Hall (1978), *Journal of Political Economy*, 86(6), 971–987.

- Hall used the Euler equation to obtain a prediction that could be tested with consumption data.
- Hall could test the model by asking whether past income predicts consumption changes, without measuring households' expectations of future income.

Reading: Cochrane (2024), *Bob Hall and Consumption*.

The intuition: consumption responds to news

- **Expected income changes are already in the plan.** An anticipated payday brings cash, but no news. Unexpected news of a permanent pay rise changes the plan now.
- **The "random walk" prediction:** information available today should not systematically predict tomorrow's consumption change.
- **Hall's evidence:** substantial support, but stock prices had some predictive power. This was not definitive proof of the PIH.

An extension beyond our two-period model

Infinite horizon, uncertainty, rational expectations and quadratic utility; a constant interest rate equal to the discount rate.

Next: evidence that motivates borrowing constraints and other departures from this benchmark.

Sources: Hall (1978); Cochrane (2024).

Empirical failure of the PIH

- Multi-period versions of the PIH generate MPCs out of transitory income shocks that are way too small relative to data
- Can show: model MPC $\approx r$ so something like 0.02 or 0.05 per year
- In contrast, empirical MPCs are around 0.3 per year on average (i.e. people consume 30 cents for every \$1)
- ... lots of heterogeneity in MPCs
- ... and this heterogeneity depends on things like liquid wealth, i.e. more cash-strapped households have higher MPCs
- See evidence on next slides
- So perhaps Keynes wasn't as wrong as we thought?

Empirical failure of the PIH (from EC1B1 notes)

Measuring the MPC

- Parker, Souleles, Johnson, McClelland (2012):
  - Look at 2008 Economic Stimulus Payments
  - People got checks from the government (~1000 USD)
  - Timing was random (based on last digits of SS#)
  - Compare those that got a check at time t with those that didn't (treatment vs. control)
  - How much more did treated people spend over a 3-month period

  ... Another natural experiment!

  - Randomisation means on average the treated and control people are the same

Empirical failure of the PIH (from EC1B1 notes)

Evidence of MPC Levels

TABLE 3—THE RESPONSE TO ESP RECEIPT AMONG HOUSEHOLDS RECEIVING PAYMENTS

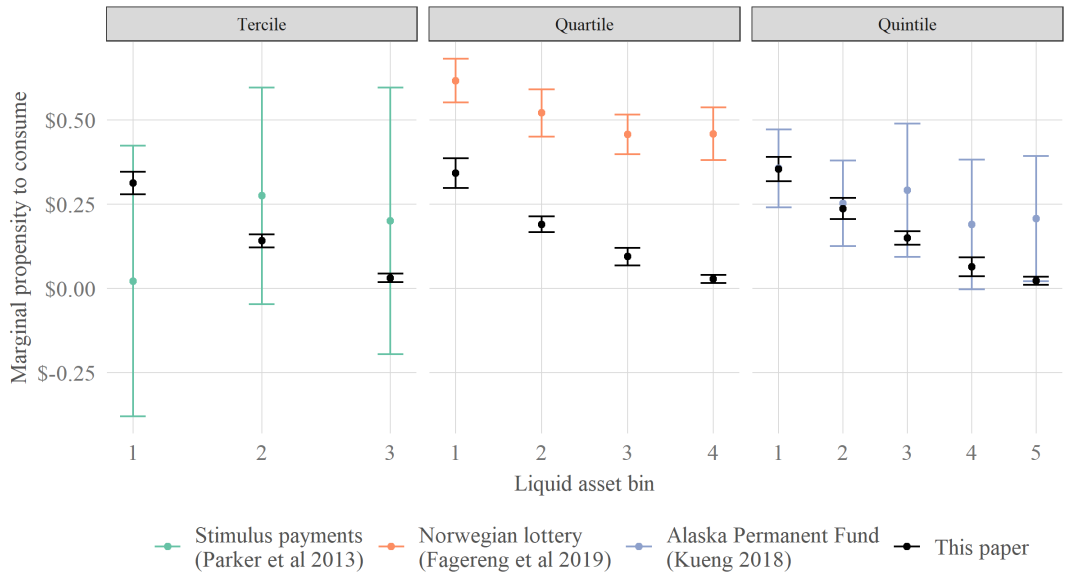
	Dollar change in		Percent change in		Dollar change in	
	Nondurable spending	All CE goods and services	Nondurable spending	All CE goods and services	Nondurable spending	All CE goods and services
	OLS	OLS	OLS	OLS	2SLS	2SLS
Panel A. Sample of all households ($N = 17{,}478$)						
ESP	0.117	0.507			0.123	0.509
	(0.060)	(0.196)			(0.081)	(0.253)
$I(ESP)$			2.63	3.97		
			(1.07)	(1.34)		
$I(ESP_{i,t} > 0 \text{ for any } t)_i$	9.58	21.21	−0.88	−1.17	8.23	20.77
	(36.07)	(104.00)	(0.50)	(0.63)	(38.79)	(112.18)
Panel B. Sample of households receiving ESPs ($N = 11{,}239$)						
ESP	0.185	0.683			0.252	0.866
	(0.066)	(0.219)			(0.103)	(0.329)
$I(ESP)$			3.91	5.63		
			(1.33)	(1.69)		
Panel C. Sample of households receiving only on-time ESPs ($N = 10{,}488$)						
ESP	0.214	0.590			0.308	0.911
	(0.070)	(0.217)			(0.112)	(0.342)
$I(ESP)$			4.52	6.05		
			(1.50)	(1.89)		

- Technical point:
  - These estimates come from regressions
  - Cf. your metrics class!

Source: Parker, Souleles, Johnson, McClelland (2012)

Empirical failure of the PIH

Figure 6: Marginal Propensity to Consume by Asset Buffer



Note: This figure compares the estimates of heterogeneity by assets in the passthrough of income shocks to consumption. Parker et al. (2013), Fagereng, Holm and Natvik (2018) and Kueng (2018) use terciles, quartiles, and quintiles respectively. To enable comparability with these prior papers, we calculate the marginal propensity to consume (instead of the elasticity of consumption to income) using their respective bin cutoffs. Our paper, Parker et al. (2013), and Kueng (2018) measure the MPC on nondurables. Fagereng, Holm and Natvik (2018) measures the MPC on total consumption. See Section 3.5 for details.

Source: Ganong-Jones-Noel-Farrell-Greig-Wheat

How to fix the PIH prediction?

Huge literature. Most common fixes:

- Credit constraints, i.e. drop assumption that can borrow as much as you want at rate r
- Behavioral theories
  - mental accounting
  - present bias or "salience of the present" $\Rightarrow$ dynamic inconsistency and procrastination
  - ...

Consumption, saving and interest rates in general equilibrium

Consumption, saving, interest rates in general equilibrium

- So far: exogenously given interest rate r
- Now: equilibrium determination of r
- Continue to work with two-period model
- Economy without production, i.e. endowment economy
- Main application: representative household model
- But start with environment with multiple households $i = 1, ..., I$ for reason that will become clear shortly

Borrowing and saving in equilibrium

- I households $i = 1, ..., I$
- Recall: $a_i > 0$ = saving, $a_i < 0$ = borrowing
- Claim: in a closed system, e.g. a closed economy or this classroom

$$\sum_{i=1}^{I} a_i = 0$$

- Logic: there is a saver for every borrower, i.e. you can only borrow if someone takes the other side of this trade and lends to you
- What if all households are alike, i.e. $a_i = a$ for all $i = 1, ..., I$? Then clearly

$$a_i = 0 \quad \text{all } i = 1, ..., I$$

  i.e. in equilibrium noone will borrow and noone will save
- Next: competitive equilibrium with rep household $\Rightarrow$ exactly what happens

Competitive equilibrium with representative household

Definition: a competitive equilibrium are quantities (c_1, c_2, a) and an interest rate r such that

1. Utility maximization: taking as given r, the representative household chooses (c_1, c_2, a) to solve

$$\max_{c_1, c_2, a} \; u(c_1) + \beta u(c_2) \quad \text{s.t.} \quad c_1 + a = y_1, \quad c_2 = y_2 + (1 + r)a$$

2. Profit maximization: none because no production
3. Market clearing: demand = supply for goods and credit market

$$\begin{aligned} \text{goods in period 1:} \quad & c_1 = y_1 \\ \text{goods in period 2:} \quad & c_2 = y_2 \\ \text{credit market:} \quad & a = 0 \end{aligned}$$

Note on goods market: if no borrowing, can only consume your income

What's the point of this equilibrium model?

- We usually solve for equilibrium quantities like (c_1, c_2, a) because we're interested in their behavior, how they change with parameters etc
- But here we already know that $(c_1, c_2, a) = (y_1, y_2, 0)$ so what's even the point?
- Answer: the model makes interesting predictions about the equilibrium interest rate

Solving for the equilibrium interest rate

- Household maximization $\Rightarrow$ Euler equation

$$u'(c_1) = \beta(1 + r)u'(c_2)$$

- But we know that $c_1 = y_1$ and $c_2 = y_2$. Therefore

$$1 + r^* = \frac{1}{\beta}\frac{u'(y_1)}{u'(y_2)}$$

- Parametric example from earlier: $u'(c) = c^{-\sigma}$

$$1 + r^* = \frac{1}{\beta}\left(\frac{y_2}{y_1}\right)^{\sigma}$$

- Note: for entire economy, reasonable to assume $y_1 \approx y_2 \Rightarrow 1 + r^* \approx 1/\beta$

Comparative statics: how does r^* depend on parameters?

- Recall

$$1 + r^* = \frac{1}{\beta}\left(\frac{y_2}{y_1}\right)^{\sigma}$$

- What happens to equilibrium interest rate r^* when
  - $\beta \uparrow$?
  - $y_1 \uparrow$?
  - $y_2 \uparrow$?

  and what is the intuition?

- Note: questions like these are excellent exam questions!

Comparative statics: answers and intuition

$$r^* = \frac{1}{\beta}\left(\frac{y_2}{y_1}\right)^{\sigma} - 1, \qquad \sigma > 0.$$

Holding other parameters fixed:

- Higher β lowers r^*. More patient households want to save more at the initial interest rate. The rate must fall to discourage saving and clear the credit market.
- Higher y_1 lowers r^*. With more income today, households want to save some of the increase to smooth consumption. The rate must fall to offset this extra desired saving.
- Higher y_2 raises r^*. With more income tomorrow, households want to borrow against it to consume more today. The rate must rise to discourage borrowing.

In equilibrium, saving remains zero: the interest rate adjusts because everyone cannot save or borrow simultaneously.

Introducing uninsurable income risk

- So far, future income was known when households chose how much to save.
- Now: future income differs across households and is uncertain when they make their saving decisions.
- Keep two periods, no production, and a bond in zero net supply.
- Questions: How does uninsurable income risk affect desired saving and the equilibrium interest rate?
- A two-period Huggett benchmark: a simple way to understand incomplete markets, not a quantitatively realistic model.

Idiosyncratic income risk

- A unit mass of ex ante identical households: same preferences, current income $y_1 > 0$, and future-income distribution.
- Each household's future income is

$$y_2 = \begin{cases} y_{2L} = \bar{y}_2 - \varepsilon & \text{with probability } 1/2, \\ y_{2H} = \bar{y}_2 + \varepsilon & \text{with probability } 1/2, \end{cases} \qquad 0 \leq \varepsilon < \bar{y}_2.$$

- Mean income: $\mathbb{E}[y_2] = \bar{y}_2$. More risk: raise ε while holding $\bar{y}_2$ fixed.
- In the population, half receive each realization. Aggregate future income is $\bar{y}_2$: no aggregate uncertainty.
- Same prospects before the shock; different incomes after it.

An incomplete financial market

- Only a one-period risk-free bond. Its gross return is $R = 1 + r > 0$.
- As before, a is saving today; its future payoff is Ra:

$$c_1 = y_1 - a, \qquad c_{2L} = y_{2L} + Ra, \qquad c_{2H} = y_{2H} + Ra.$$

- $a > 0$: lending; $a < 0$: borrowing. The bond has zero net supply.
- No state-contingent claims: households cannot buy insurance that pays only when their own income is low.
- The consumption good cannot be stored; the bond transfers resources between households.

The household problem

Taking R as given, choose saving before the income shock:

$$\max_a \left\{ u(\underbrace{y_1 - a}_{c_1}) + \beta \left[\frac{1}{2} u(\underbrace{y_{2L} + Ra}_{c_{2L}}) + \frac{1}{2} u(\underbrace{y_{2H} + Ra}_{c_{2H}}) \right] \right\}.$$

- Use the same CRRA preferences as earlier:

$$u(c) = \frac{c^{1-\sigma} - 1}{1 - \sigma}, \quad u'(c) = c^{-\sigma}, \quad \sigma > 0,$$

  with $u(c) = \log c$ when $\sigma = 1$.

- Positive consumption in every state requires

$$y_1 - a > 0, \quad y_{2L} + Ra > 0 \quad \Rightarrow \quad \boxed{-\frac{y_{2L}}{R} < a < y_1.}$$

- $-y_{2L}/R$ is the natural borrowing limit: debt must be repayable even with low income.

The Euler equation under uncertainty

From now on, assume $\sigma = 1$: $u(c) = \log c$, $u'(c) = 1/c$.

- Differentiate expected utility with respect to saving:

$$-u'(c_1) + \beta R \left[\frac{1}{2} u'(c_{2L}) + \frac{1}{2} u'(c_{2H}) \right] = 0.$$

- Therefore,

$$\boxed{u'(c_1) = \beta R \mathbb{E}[u'(c_2)].}$$

- Saving one more unit costs $u'(c_1)$ today and yields R units tomorrow, valued at expected marginal utility.

- With log utility:

$$\frac{1}{c_1} = \frac{\beta R}{2} \left(\frac{1}{c_{2L}} + \frac{1}{c_{2H}} \right).$$

- In general, $\mathbb{E}[u'(c_2)] \neq u'(\mathbb{E}[c_2])$: the distribution matters.

Competitive equilibrium with income risk

An equilibrium consists of R^* and $(a^*, c_1^*, c_{2L}^*, c_{2H}^*)$ such that:

1. Household optimization: choices solve the household problem taking R^* as given.

2. Bond-market clearing:

$$\int a_i^* \, di = 0.$$

   Identical households choose the same saving before the shock, so $a^* = 0$.

3. Goods-market clearing:

$$c_1^* = y_1, \qquad \frac{1}{2}c_{2L}^* + \frac{1}{2}c_{2H}^* = \frac{1}{2}y_{2L} + \frac{1}{2}y_{2H}.$$

Individual budgets and $a^* = 0$ imply $c_{2L}^* = y_{2L}$, $c_{2H}^* = y_{2H}$.

Income risk passes fully into consumption risk in this benchmark.

The equilibrium interest rate under income risk

- Substitute $a^* = 0$, $c_1^* = y_1$, and $c_2^* = y_2$ into the Euler equation:

$$u'(y_1) = \beta R^* \mathbb{E}[u'(y_2)].$$

- Hence

$$\boxed{R^* = \frac{u'(y_1)}{\beta \mathbb{E}[u'(y_2)]}.}$$

- With log utility and two equally likely states:

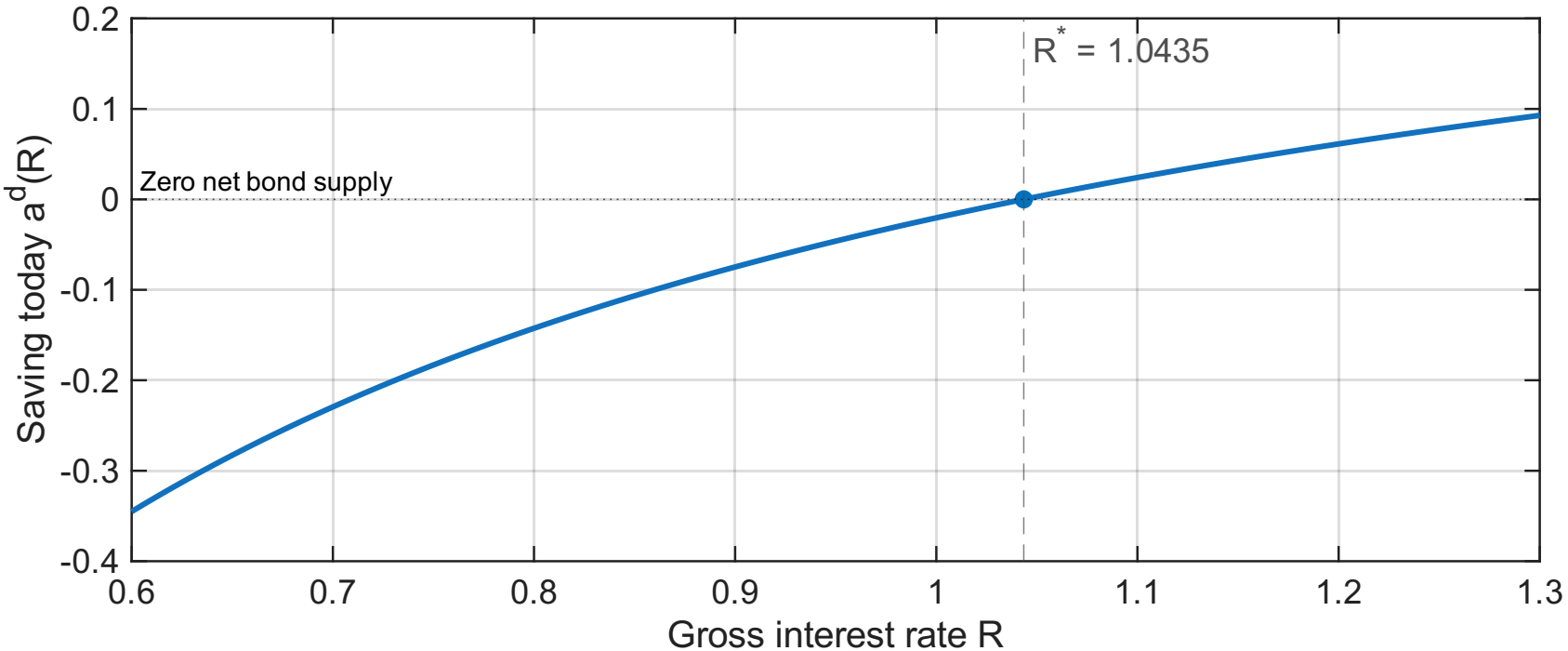
$$R^* = \frac{1/y_1}{\frac{\beta}{2}\left(1/y_{2L} + 1/y_{2H}\right)} = \boxed{\frac{\bar{y}_2^2 - \varepsilon^2}{\beta y_1 \bar{y}_2}}.$$

- If $\varepsilon = 0$, recover our earlier deterministic result:

$$R^* = \frac{\bar{y}_2}{\beta y_1}.$$

Household asset demand: numerical example

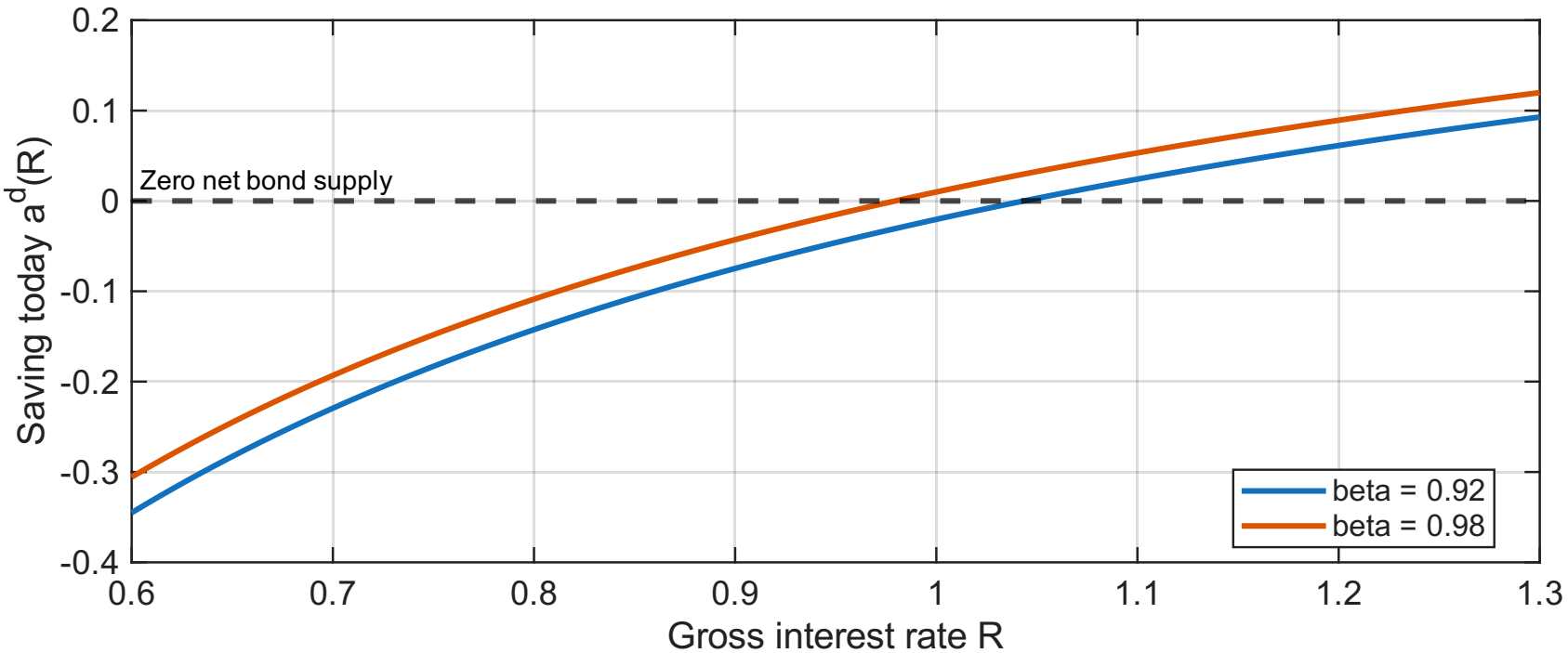
$\beta = 0.92$, $\sigma = 1$, $y_1 = 1$, $y_2 \in \{0.8, 1.2\}$, each with probability $1/2$.



- At $R = 1$, households want to borrow: $a^d(1) < 0$.
- Equilibrium: $R^* \approx 1.043$, $r^* \approx 4.35\%$ per period, $a^* = 0$.

Asset demand: higher patience

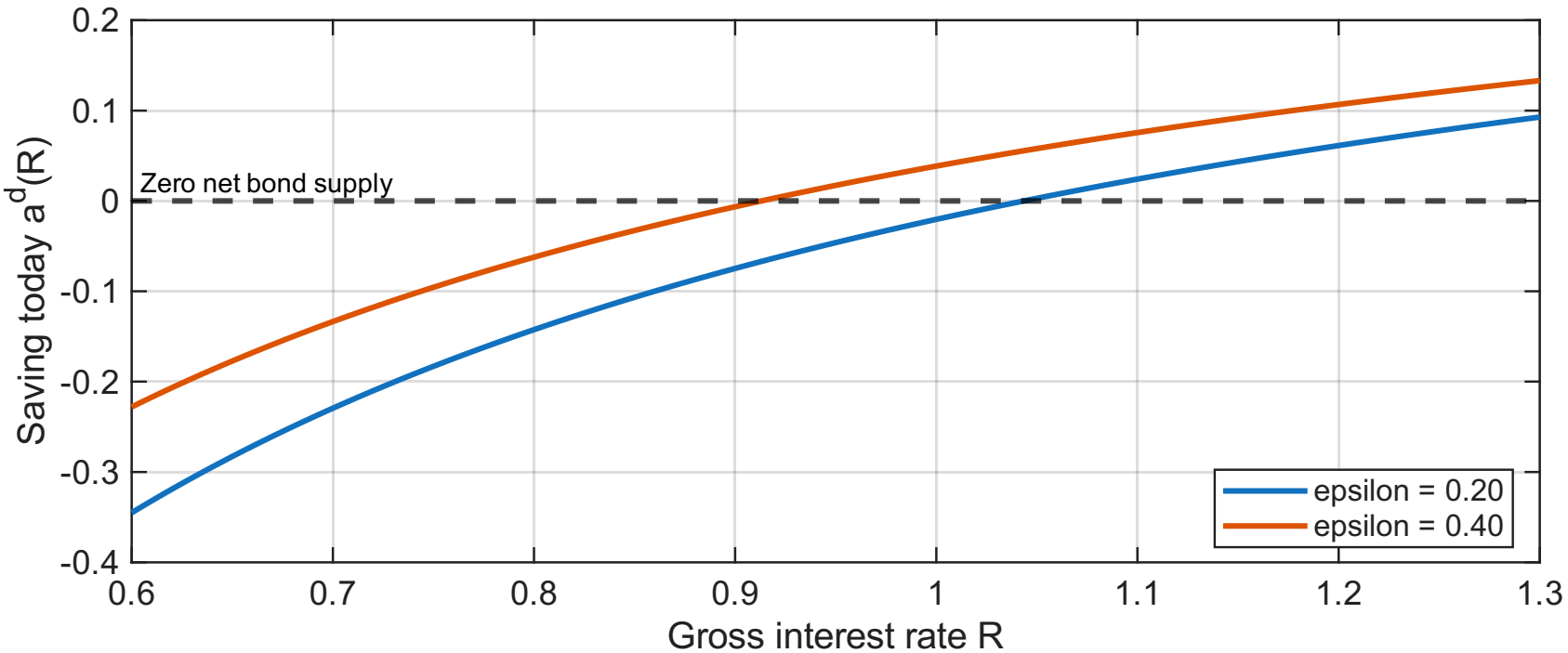
Raise β from 0.92 to 0.98; keep incomes, probabilities and σ fixed.



- Higher β: more saving at each R, so $a^d(R)$ shifts up.
- Market clearing: R^* falls from about 1.043 to 0.980.

Asset demand: higher income risk

Raise ε from 0.2 to 0.4; keep $\bar{y}_2 = 1$, $\beta = 0.92$ and $\sigma = 1$.



- Equally likely incomes: $\{0.8, 1.2\} \to \{0.6, 1.4\}$.
- More risk shifts $a^d(R)$ up; R^* falls from 1.043 to 0.913.

Finding equilibrium numerically

Define excess bond demand $Z(R) = a^d(R)$. Equilibrium solves $Z(R^*) = 0$.

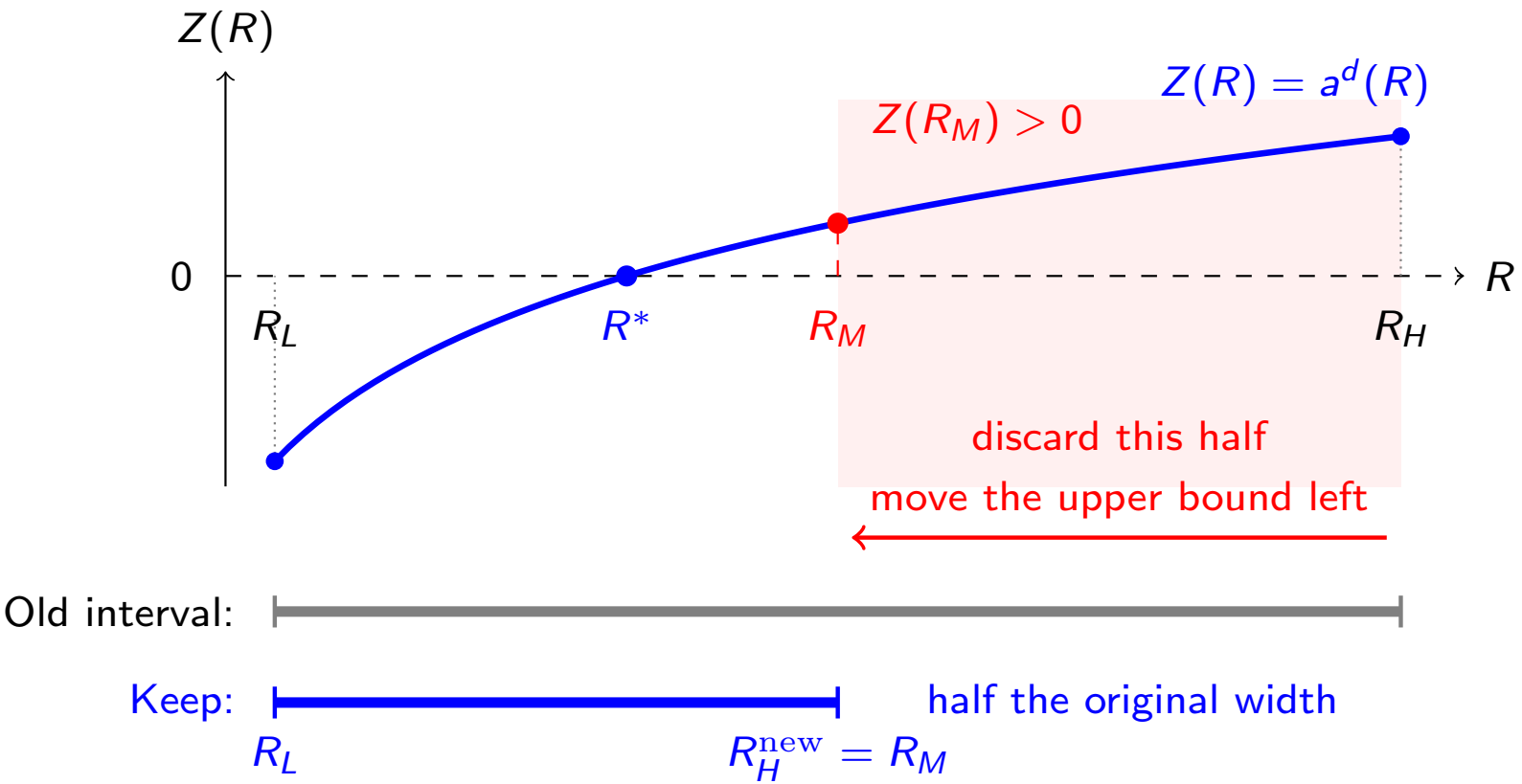
1. Choose a bracket $[R_L, R_H]$ with $Z(R_L) < 0 < Z(R_H)$.
2. Try the midpoint $R_M = (R_L + R_H)/2$.
3. Solve the household Euler equation at R_M to find $a^d(R_M)$.
4. If $a^d(R_M) > 0$, desired saving is too high: lower the upper endpoint to R_M. If $a^d(R_M) < 0$, raise the lower endpoint to R_M.
5. Repeat until excess demand is sufficiently close to zero.

MATLAB implementation: Run `lecture3_extra.m`. `huggett_household.m` solves the Euler equation using `fzero`; `huggett_equilibrium.m` bisects R on $[0.2, 3]$ until $|a| < 10^{-10}$ (at most 100 iterations).

Numerical and closed-form R^* agree to within 10^{-8}.

Bisection: keep the half containing the root

First midpoint: $Z(R_M) > 0$, so households want to save.



The root stays inside: $[R_L, R_H] \longrightarrow [R_L, R_M]$.

Your turn!

Have a play with the MATLAB code: change one parameter at a time, predict the shift, and see what happens!

Precautionary saving lowers the interest rate

- Prudence: $u'''(c) > 0$, so marginal utility $u'(c)$ is convex. CRRA has this property for every $\sigma > 0$.

- At fixed mean income, a mean-preserving spread raises expected marginal utility. Relative to certainty, Jensen's inequality gives

$$\tfrac{1}{2}u'(\bar{y}_2 - \varepsilon) + \tfrac{1}{2}u'(\bar{y}_2 + \varepsilon) > u'(\bar{y}_2), \qquad \varepsilon > 0.$$

- At a given R, higher risk also raises $\mathbb{E}[u'(y_2 + Ra)]$ at any feasible a: households want to save more.

- But bond supply is zero. The equilibrium return must fall:

$$\varepsilon \uparrow \Rightarrow \ \mathbb{E}[u'(y_2)] \uparrow \Rightarrow \ \boxed{R^* \downarrow .}$$

- Desired saving rises at the old rate; actual equilibrium saving remains $a^* = 0$.

Complete versus incomplete markets

Hold preferences and resources fixed: $y_1 = \mathbb{E}[y_2] = 1$, with $\varepsilon > 0$.

- Complete markets: state-contingent claims insure individual income risk. With no aggregate risk and identical initial positions,

$$c_1 = c_{2L} = c_{2H} = 1 \quad \Rightarrow \quad R^*_{CM} = \frac{1}{\beta}.$$

- Incomplete markets: only the risk-free bond is available,

$$c_{2L} = y_{2L}, \quad c_{2H} = y_{2H}, \qquad R^*_{IM} = \frac{u'(1)}{\beta \mathbb{E}[u'(y_2)]}.$$

- Prudence gives $\mathbb{E}[u'(y_2)] > u'(1)$, hence

$$\boxed{R^*_{IM} < R^*_{CM}.}$$

- In our numerical example, the net rate falls from 8.70% under full insurance to 4.35% without it, per model period.

A transparent log-utility example

With $u(c) = \log c$, $y_1 = 1$, and two equally likely income states:

$$R^* = \frac{1}{\frac{\beta}{2}\left[\frac{1}{\bar{y}_2 - \varepsilon} + \frac{1}{\bar{y}_2 + \varepsilon}\right]} = \boxed{\frac{\bar{y}_2^2 - \varepsilon^2}{\beta \bar{y}_2}}, \qquad 0 \le \varepsilon < \bar{y}_2.$$

- More risk at a fixed mean lowers the rate:

$$\frac{\partial R^*}{\partial \varepsilon} = -\frac{2\varepsilon}{\beta \bar{y}_2} < 0 \quad (\varepsilon > 0).$$

- Higher mean income at fixed risk raises the rate:

$$\frac{\partial R^*}{\partial \bar{y}_2} = \frac{1}{\beta}\left(1 + \frac{\varepsilon^2}{\bar{y}_2^2}\right) > 0.$$

- Without risk, $R^* = \bar{y}_2/\beta$: the consumption-smoothing result from earlier.

AI: two opposing channels

- Higher expected future income: raise $\bar{y}_2$ at fixed ε. Both income states improve, so

$$\mathbb{E}[u'(y_2)] \downarrow \quad \Rightarrow \quad R^* \uparrow .$$

  Households want more consumption today; the rate rises to discourage borrowing.

- Greater individual earnings risk: raise ε at fixed $\bar{y}_2$. Some workers benefit more; others face displacement:

$$\mathbb{E}[u'(y_2)] \uparrow \quad \Rightarrow \quad R^* \downarrow .$$

- Here ε describes dispersion **across households**, not aggregate uncertainty about whether AI transforms the economy. That would require a different model.

An illustrative ten-year calibration

- One model period is ten years. Set $\sigma = 1$, $y_1 = 1$, equal state probabilities, and $\beta = 0.98^{10} \approx 0.817$.
- Baseline annual income growth is 1%: $\bar{y}_2^{\text{base}} = 1.01^{10}$. An AI boom adds 15% to this future income **level**:

$$\bar{y}_2^{\text{boom}} = 1.15 \times 1.01^{10}.$$

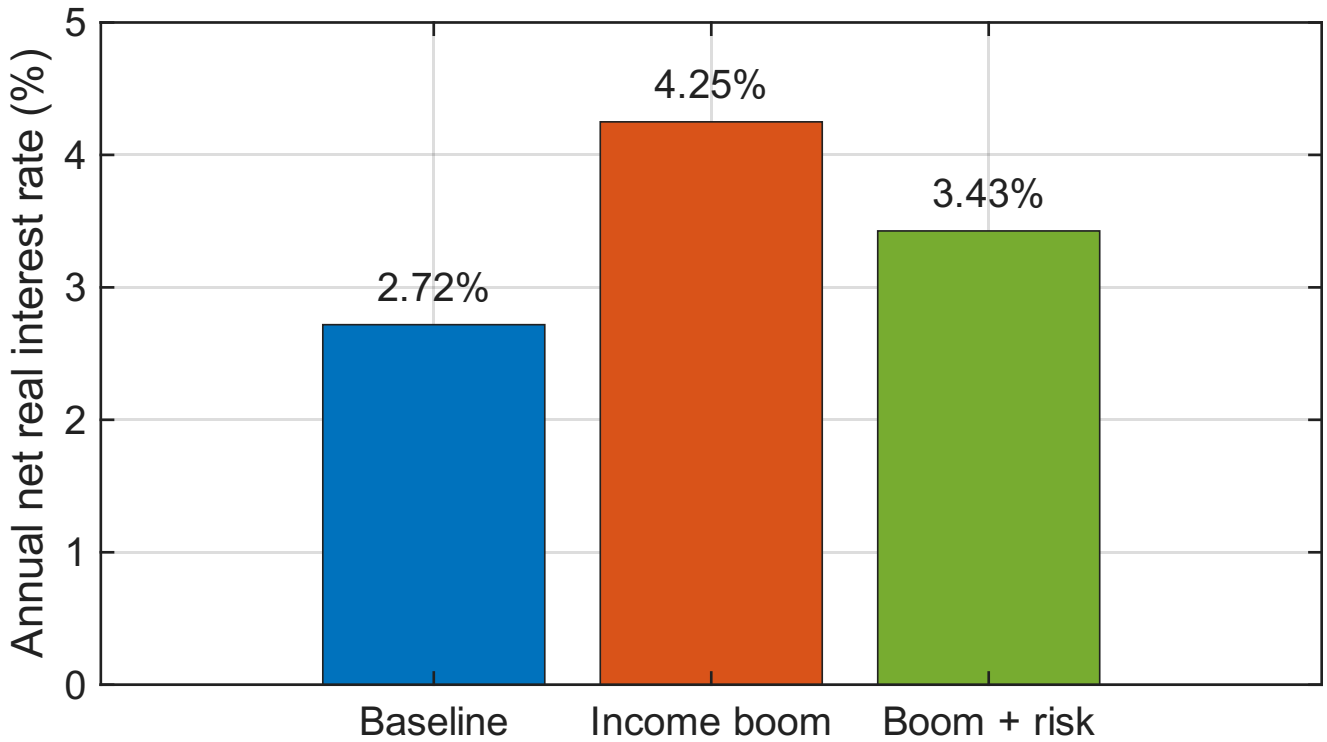
Scenario	Mean income $\bar{y}_2$	Risk ε
Baseline	1.1046	0.20
Income boom	1.2703	0.20
Boom + risk	1.2703	0.40

Convert the ten-year gross return into an annual net rate:

$$\boxed{r_{\text{ann}} = (R^*)^{1/10} - 1.}$$

Illustrative assumptions, not an empirical forecast of AI's effects.

AI scenarios: numerical results



- Higher expected income raises the annual real rate from 2.72% to 4.25%.
- More individual earnings risk lowers it to 3.43%: a partial offset, leaving the rate above baseline.
- In all three equilibria, aggregate bond holdings remain zero.

Main lessons from the Huggett benchmark

1. Insurance matters. With prudent preferences, uninsurable income risk raises precautionary-saving demand and lowers the equilibrium risk-free rate relative to full insurance.

2. Expected income matters. Higher future income in every state encourages borrowing today, raising the equilibrium rate.

3. Risk matters. Greater individual earnings risk at a fixed mean lowers the equilibrium rate, offsetting the expected-income channel in the AI example.

The rate adjusts while $a^* = 0$. This two-period, pure-exchange benchmark has no capital accumulation, aggregate risk, or initial household heterogeneity.

From interest rates to stock prices

- So far: the Euler equation determines the equilibrium return on a risk-free bond.
- A share is a claim to future dividends. How much should we pay for it today?
- The same consumption-saving logic prices both bonds and shares.
- New ingredient: dividends may be low precisely when consumption is low.

We now move from individual income risk to aggregate risk: the whole economy can have a good or a bad outcome.

A simple economy with a dividend-paying tree

- Two dates, $t = 1, 2$; one consumption good; one representative household representing identical households.
- Today: endowment $y_1 > 0$ and ownership of one tree.
- Tomorrow: the tree pays $d_{2L} > 0$ or $d_{2H} > 0$, each with probability $1/2$, then expires.
- The tree has one divisible share, traded today at price p.
- A risk-free bond pays gross return $R = 1 + r$ and has zero net supply. Goods cannot be stored.

The tree is a simple firm with given dividends. Its value is endogenous; its dividends are exogenous.

The household chooses bonds and shares

Taking p and R as given, choose saving a and shares θ to maximise

$$u(c_1) + \beta \left[\frac{1}{2} u(c_{2L}) + \frac{1}{2} u(c_{2H}) \right]$$

subject to

$$c_1 + a + p(\theta - 1) = y_1,$$
$$c_{2s} = Ra + \theta d_{2s}, \qquad s = L, H.$$

Use CRRA preferences, $\sigma > 0$, and $0 < \beta < 1$.

- $a > 0$: lend; $a < 0$: borrow.
- $\theta - 1$ is the number of shares bought beyond the initial holding.
- Buying additional shares reduces current consumption; selling shares releases resources today.
- Choices must leave consumption positive in every state.

Two Euler equations, the same economic logic

Substitute the budgets and differentiate with respect to each asset holding.

For bonds:

$$u'(c_1) = \beta R \mathbb{E}[u'(c_2)].$$

For shares:

$$p\, u'(c_1) = \beta \mathbb{E}[u'(c_2) d_2].$$

- One more share costs p units today and pays d_{2s} units tomorrow in state s.
- At the optimum, the marginal utility cost equals the expected discounted marginal utility benefit.

For shares, marginal utility and the dividend are inside the same expectation.

Competitive equilibrium: quantities first, prices next

An equilibrium consists of prices p, R and quantities $a, \theta, c_1, c_{2L}, c_{2H}$ such that the household optimises at those prices and markets clear.

Market clearing:

$$a = 0, \qquad \theta = 1, \qquad c_1 = y_1, \qquad c_{2s} = d_{2s}.$$

Substitute these quantities into the Euler equations:

$$R = \frac{u'(y_1)}{\beta \mathbb{E}[u'(d_2)]},$$

$$p = \beta \frac{\mathbb{E}[u'(d_2) d_2]}{u'(y_1)}.$$

The household chooses its portfolio. Prices adjust so that holding the whole tree and zero bonds is optimal.

What makes a future payment valuable?

Define the stochastic discount factor:

$$m_{2s} = \beta \frac{u'(c_{2s})}{u'(c_1)}.$$

The pricing equations become

$$p = \mathbb{E}[m_2 d_2], \qquad R = \frac{1}{\mathbb{E}[m_2]}.$$

- m_{2s} converts a future payment into current-consumption units, with a weight depending on the state.
- Low consumption tomorrow means high marginal utility and a high valuation of an extra payment.
- Under certainty, $m_2 = 1/R$, so the familiar formula is $p = d_2/R$.

A payment is valuable because of both how much it delivers and when it arrives.

A numerical example: a bond and a stock

$$u(c) = \log c, \quad \beta = 0.92, \quad y_1 = 1, \quad d_{2L} = 0.8, \quad d_{2H} = 1.2.$$

State (probability)	Consumption c_{2s}	Discount factor m_{2s}	Stock payoff d_{2s}
Low (1/2)	0.8	1.1500	0.8
High (1/2)	1.2	0.7667	1.2

Using exact values:

$$p = \tfrac{1}{2}(1.15)(0.8) + \tfrac{1}{2}(0.92/1.2)(1.2) = 0.92.$$

$$R - 1 = 4.35\%, \qquad \mathbb{E}[R_2^e] - 1 = 8.70\%, \qquad R_{2s}^e = d_{2s}/p.$$

Expected equity premium: 4.35 percentage points per model period.

These are aggregate states, not individual income draws. The example is illustrative.

Why do risky assets offer a premium?

$$p = \frac{\mathbb{E}[d_2]}{R} + \mathrm{Cov}(m_2, d_2).$$

- The first term discounts expected dividends at the risk-free rate.
- Our stock pays less when consumption is low and m_2 is high, so the covariance is negative.
- Its price is therefore below $\mathbb{E}[d_2]/R$; its expected return is above R.
- A payoff concentrated in bad times provides insurance and can command a higher price and lower expected return.

Risk is about paying poorly when resources are most valuable, not just about payoff volatility.

The Lucas tree: an asset that lives forever

Now let the tree pay a positive dividend d_t every period, for $t = 0, 1, 2, \ldots$. The dividends are the economy's only goods; goods cannot be stored.

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t).$$

Shares trade after observing current dividends. θ_t is the shareholding at the start of the period; p_t is the price after the current dividend has been paid.

$$c_t + p_t \theta_{t+1} = (p_t + d_t)\theta_t.$$

- For this extension, display only equity; the bond can be priced separately by the same Euler equation.
- One outstanding share, initially owned by the household: market clearing gives $\theta_t = \theta_{t+1} = 1$ and $c_t = d_t$.

Dividends plus resale value

One more share costs p_t today. Tomorrow it delivers the dividend and a share worth p_{t+1}.

$$p_t u'(c_t) = \beta \mathbb{E}_t \left[u'(c_{t+1})(d_{t+1} + p_{t+1}) \right].$$

With equilibrium consumption $c_t = d_t$:

$$p_t = \mathbb{E}_t \left[m_{t+1}(d_{t+1} + p_{t+1}) \right],$$

$$m_{t+1} = \beta \frac{u'(d_{t+1})}{u'(d_t)}.$$

- $\mathbb{E}_t$ means expectation conditional on information available today.
- Price reflects expected future payments and how households value them across dates and states.
- With one share and no corporate debt, p_t is also the equity value of the whole firm, after today's dividend.

Source: Altug and Labadie (2008), Chapter 8, pp. 162–173.

Further reading: the value of future dividends

Substituting the pricing equation forward repeatedly expresses price in terms of future dividends and a terminal resale value.

Define the multi-period discount factor:

$$M_{t,t+j} = \beta^j \frac{u'(c_{t+j})}{u'(c_t)}.$$

Impose the no-bubble condition

$$\lim_{T \to \infty} \mathbb{E}_t[M_{t,t+T} p_{t+T}] = 0.$$

Then

$$p_t = \mathbb{E}_t \left[\sum_{j=1}^{\infty} M_{t,t+j} d_{t+j} \right].$$

Fundamental firm value is the present value of future dividends, discounted for both time and risk.

Further reading: a Lucas model we can solve exactly

With log utility and $c_t = d_t$, the pricing equation becomes

$$p_t = \beta d_t \mathbb{E}_t \left[1 + \frac{p_{t+1}}{d_{t+1}} \right].$$

Try a constant price-dividend ratio, $p_t = k d_t$. Substitution gives

$$k = \beta(1 + k) \quad \Longrightarrow \quad k = \frac{\beta}{1 - \beta}.$$

Hence the fundamental price is

$$p_t = \frac{\beta}{1 - \beta} d_t.$$

- Greater patience raises the price-dividend ratio; for $\beta = 0.92$, it is 11.5.
- This particularly simple result relies on log utility and consumption equalling the tree's dividend.

Next few lectures

- More interesting dynamic macro models with non-degenerate predictions for the quantities?
- $\Rightarrow$ add production and capital accumulation
- Use them to think about business cycles and macroeconomic stabilization policies, in particular monetary and fiscal policy
  - microfounded analogue of Keynesian IS-MP-PC model